Data Quality Assessment Report

Dataset	Issues Found	Date	Analyst
50 Patient Records	**6 Violations**	**Q2 2025**	**Junior Developer**

Dataset Preview

The table below shows the sample records provided. Cells highlighted in red contain the specific data quality violations identified in this assessment.

PatientID	PatientName	PhoneNumber	AppointmentDate	DoctorName	PaymentStatus
P001	Kwame Mensah	0244123456	2025-10-15	Dr. Osei	Paid
P002	Ama Serwa	**244789012**	**15/10/2025**	**dr. osei**	paid
P001	Kwame Mensah	0244123456	2025-10-20	Dr. Adjei	Pending
P003	Kofi Annan	0244567890	**10/16/2025**	Dr. Osei	Failed
P004	***(missing)***	0555234567	2025-10-17	Dr. Mensah	Paid
P002	Ama Serwa	**0244789012**	2025-10-15	Dr. Osei	Paid

Key: Red cells indicate a data quality violation. Hover over the table or reference Task 1 below for the exact dimension violated per cell.

Task 1 — Data Quality Dimension Violations

Each of the six standard data quality dimensions has at least one violation in the dataset. The table below documents each violation with a specific row reference and explanation.

Dimension	Record	Violation Example
Accuracy	*Row 4 (P003)*	P003: AppointmentDate listed as 10/16/2025 (October 16). No month 16 exists — this is a factually impossible date.
Completeness	*Row 5 (P004)*	P004: PatientName field is entirely empty. A patient record with no name cannot be used for SMS reminders, billing, or clinical lookup.
Consistency	*Rows 1 vs 2*	DoctorName recorded as 'Dr. Osei' in row 1 but 'dr. osei' (lowercase) in row 2. PaymentStatus is 'Paid' vs 'paid'. Same entity, different representations.
Timeliness	*Row 2 (P002)*	Row 2 (P002): AppointmentDate entered as '15/10/2025' (DD/MM/YYYY) while the system standard is YYYY-MM-DD. If parsed literally by software, 15 is treated as the month — making it appear as a future date in month 15, which is invalid and stale.
Validity	*Row 2 (P002)*	P002 (first entry): PhoneNumber is '244789012' — missing the leading zero required by Ghana's mobile number format (10 digits starting with 0). The number is structurally invalid for SMS delivery.
Uniqueness	*Rows 2 & 6*	P002 (Ama Serwa) appears twice with identical AppointmentDate (2025-10-15) and DoctorName (Dr. Osei). This is a duplicate record that will inflate patient counts and trigger double billing.

Task 2 — Business Impact Assessment

Each data quality issue creates downstream operational, financial, or clinical problems. The table below maps each violation to its specific business consequence and the function most affected.

Dimension	Business Problem	Affected Function
Accuracy	An impossible date (month 16) will cause appointment scheduling software to throw errors or silently discard the record. The appointment will not appear in daily schedules, the patient will not receive a reminder, and the doctor's calendar will be wrong.	**Operations / Clinical**
Completeness	Without a patient name, the SMS system cannot personalise or even address the reminder message. Billing cannot generate an invoice. Clinical staff cannot identify the patient at the front desk.	**Operations / Finance / Clinical**
Consistency	If the system groups appointments by DoctorName, 'Dr. Osei' and 'dr. osei' are treated as two different doctors. Reports will under-count one doctor's caseload. PaymentStatus inconsistencies break financial roll-up queries.	**Finance / Operations**
Timeliness	A date in DD/MM/YYYY format parsed by a YYYY-MM-DD system results in either an error or a wildly incorrect date. Automated reminders fire at wrong times or not at all. Reports misplace the appointment in the timeline.	**Operations**
Validity	An invalid phone number (missing leading zero) means the SMS gateway will reject the message. The patient receives no reminder and is likely to miss the appointment — a direct revenue and care-quality loss.	**Operations**
Uniqueness	The duplicate P002 record causes the patient count to be inflated by 1. If both records trigger billing, the patient is charged twice. Reports show a higher appointment volume than actual, skewing capacity planning.	**Finance / Operations**

Task 3 — Recommended Solutions (Top 3 Critical Issues)

The three issues below were selected as highest priority based on their direct, immediate impact on MedTrack Ghana's core operations: SMS delivery, revenue integrity, and reporting accuracy.

01 [Validity] Phone Number Format Validation

Why critical: This is the most direct cause of SMS failures — the core operational product of MedTrack Ghana.

Technical Solution

Implement a server-side validation rule: PhoneNumber must match the regex ^0[235][0-9]{8}$ (Ghanaian mobile numbers: 10 digits, starting with 02x, 03x, or 05x). Reject or flag any record that does not conform at the point of data entry.

Responsibility

Junior Developer (backend input validation) + Database Administrator (constraint on the phone_number column).

Verification

Run a query: SELECT COUNT(*) FROM appointments WHERE phone_number NOT REGEXP '^0[235][0-9]{8}$' — count should return 0. Conduct a test SMS blast to 10 records and confirm 100% delivery.

02 [Uniqueness] Duplicate Record Detection & Prevention

Why critical: Double-billing patients directly damages MedTrack's financial integrity and patient trust — a critical risk for a health startup.

Technical Solution

Add a composite UNIQUE constraint on (PatientID, AppointmentDate, DoctorName). Write a deduplication script that identifies existing duplicates (same PatientID + AppointmentDate + DoctorName), retains the most recent record, and archives the other.

Responsibility

Database Administrator (constraint) + Junior Developer (deduplication script) + QA team (validation).

Verification

After deduplication, run: SELECT PatientID, AppointmentDate, DoctorName, COUNT(*) FROM appointments GROUP BY PatientID, AppointmentDate, DoctorName HAVING COUNT(*) > 1 — should return zero rows.

03 [Consistency] Data Standardisation (Case & Date Format)

Why critical: Inconsistent casing breaks grouping in all reports and inconsistent date formats break every time-based operation.

Technical Solution

Apply two transformations: (1) Normalise DoctorName and PaymentStatus to Title Case and Proper Case respectively using an ETL script (e.g., Python with .title() / .capitalize()). (2) Enforce ISO 8601 (YYYY-MM-DD) as the only accepted date format — add a CHECK constraint and a pre-insert parser that attempts to convert DD/MM/YYYY and MM/DD/YYYY inputs automatically before rejection.

Responsibility

Junior Developer (ETL script + parser) + Data Analyst (validation of report outputs) + Team Lead (sign-off).

Verification

Query: SELECT DISTINCT DoctorName FROM appointments — all entries should follow 'Dr. Firstname Lastname' pattern. Query: SELECT COUNT(*) FROM appointments WHERE AppointmentDate NOT REGEXP '^[0-9]{4}-[0-9]{2}-[0-9]{2}$' — should return 0.

Task 4 — Biggest Risk of Poor Data Consistency (Developer Perspective)

Perspective: Junior Developer / Software Engineering

From a software engineering perspective, the biggest risk of poor data consistency is silent system divergence — where different parts of the application operate on contradictory versions of the same truth, producing results that are individually plausible but collectively wrong.

Consider the 'dr. osei' vs 'Dr. Osei' inconsistency. The SMS module, which queries by DoctorName to build a schedule, will build two separate appointment lists for what is actually one doctor. Neither list throws an error. Both look valid. But the doctor's dashboard shows incomplete appointments, patients assigned to 'dr. osei' receive no reminder, and capacity reports show two doctors where there is one.

This is especially dangerous in health tech because:

- The system will not crash — it will silently serve wrong data, making the problem hard to detect.
- Downstream code (billing, analytics, clinical alerts) inherits the inconsistency and amplifies it with each new feature built on top.

- In a medical context, a missed appointment caused by a name mismatch is not just a UX failure — it is a patient safety event.

The technical solution is not only standardisation at the database level, but enforcing a single source of truth through referential integrity — storing DoctorName as a foreign key to a Doctors table, not as a free-text field. Consistency cannot be achieved through convention alone; it must be enforced structurally.